Critical Digital Humanities in Generative AI: Enhancing Critical Thinking in Education

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Introduction

This proposal outlines a research project at the intersection of Critical Digital Humanities (CDH), critical thinking, and generative artificial intelligence (AI) to address the educational opportunities and challenges these technologies present within the humanities. Adopting a CDH perspective that emphasises human experience and critical thinking while engaging with technological artefacts, this proposal critiques the subjective influence of generative AI (GenAI) tools on educational practices. GenAI technologies have been recognised for their innovative possibilities for knowledge acquisition, teaching, and research (İpek et al. 2023; Lo 2023). Research studies have highlighted their varying benefits across different subject domains, aiding educators in curriculum development and serving students as virtual tutors, such as in information technology education (Tian et al. 2023). However, these tools also raise significant concerns about the possible erosion of critical thinking skills in educational contexts (Michel-Villarreal et al. 2023; van den Berg / du Plessis 2023). By addressing the ethical considerations of using AI, the project aims to develop a framework ensuring these technologies enhance rather than undermine critical thinking.

Background and Context

The critical turn in Digital Humanities (DH), introduced by Liu (2012), has called for a more activist role that confronts social issues through cultural critique, particularly regarding gender, race, ideology, and politics (Gold / Klein 2019). This movement advocates for the discipline to take a political stance on contemporary events, thus uniting researchers at the intersection of technology and social justice (Berry 2013). In this context, Berry (2022) further argued that to effectively critique digital capitalism and technological transformation, Critical Digital Humanities (CDH) should integrate critical theory. Agreeing with Berry on the need to critique technology, Casani (2018) proposed a 'critical' interpretation of CDH that emphasises subjective perspectives, prioritising individual experiences over traditional social critique. This shift from critical theory to critical thinking allows for a nuanced understanding of how personal experiences shape our interactions with digital technologies. This perspective is particularly relevant to Generative AI (GenAI) tools, which have been critiqued for reproducing systemic biases encoded in their datasets. GenAI tools apply arbitrary ethical and argumentation frameworks, filtering and shaping their outputs. Students' increased reliance on these tools may divert critical inquiry and result in overlooking their inner functioning, thus undermining critical engagement with the information produced (Guo / Lee 2023). Moreover, reliance on AI-generated content may discourage independent thinking, leading to a decline in essential cognitive processing and analytical skills (Michel-Villarreal et al. 2023). In response to these challenges, several researchers advocate for developing guidelines, policies, and frameworks to address these limitations and biases (van den Berg / du Plessis 2023; İpek et al. 2023). Critical thinking—rooted in ancient Greek philosophy—emphasises self-reflection, rationality, and dispositions, such as open-mindedness and flexibility (McKirahan 2003; Plato 1997; Aristotle 2009). Thus, fostering critical thinking in educational settings is crucial to empower students to engage thoughtfully with AI tools and critically evaluate their implications.

Research Objectives

This proposal aims to:

- Critically analyse the educational implications of GenAI through a CDH lens, focusing on their impact on students' critical thinking skills.
- Integrate critical thinking into the design and use of GenAI tools in educational settings, utilising a mixed methods approach that combines qualitative insights from educators with quantitative assessments of student outcomes.
- Develop practical guidelines for educators that enhance critical thinking while addressing challenges such as bias and transparency, informed by feedback from pilot studies and case analyses.

Methodology

This research will employ a mixed methods approach to investigate the intersection of CDH, critical thinking, and Generative AI (GenAI) in educational settings. First, a comprehensive literature review will identify existing scholarship on CDH, critical thinking, and AI ethics while highlighting the gaps this study aims to address. Next, case studies involving GenAI tools like ChatGPT and Copilot will assess how these tools influence critical-thinking development. Data collection will include structured surveys and semi-structured interviews with educators to gather qualitative insights into their experiences using GenAI. The survey design will focus on themes such as perceived benefits and challenges related to bias. A pilot study will be conducted with a small group of educators and students to test preliminary guidelines developed from the initial findings. Feedback from this phase will inform iterative revisions of the guidelines. Finally, clear metrics for assessing improvements in students' critical thinking skills will be established through standardised assessments and qualitative analysis using established rubrics.

Expected Outcomes

This research will provide detailed insights into how GenAI technologies impact the development of critical thinking skills in educational settings, particularly within the humanities. It will develop a framework for integrating critical thinking into AI-driven teaching practices while addressing specific challenges, such as algorithmic bias and diminishing reflective engagement. The framework will offer practical solutions for educators to use GenAI tools thoughtfully. The outcomes will include improvements in students' critical thinking skills measured through pre- and post-implementation assessments and qualitative feedback from educators gathered via interviews. By addressing these challenges effectively, the research will contribute to the ethical integration of GenAI into humanities education.

Discussion and Conclusion

This research aims to demonstrate how integrating critical thinking through a CDH perspective addresses the educational challenges posed by GenAI technologies. By foregrounding ethical considerations alongside critical reflexivity, this approach would ensure that GenAI applications align with principles of openness. Critical thinking—defined as precise analysis involving thorough questioning (Dewey 1910; Facione 1990)—is essential for navigating the complexities of AI. This project emphasises fostering open-mindedness among students to analyse digital tools effectively. The findings will highlight the urgent need to address issues in AI-generated content that can obscure inherent biases. To mitigate these challenges, the research proposes implementing case-based learning methods alongside transparent guidelines for critical engagement. The proposed framework could integrate with existing Digital Humanities tools like Omeka and Scalar (Omeka n.d.; Scalar n.d.) to enhance their educational impact. By addressing these challenges effectively, this research will contribute to the ethical integration of GenAI into humanities education while fostering an environment where critical thinking can thrive.

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